

1 When Molecular and Morphological Evidence Disagree and
2 Morphology is Positively Misleading: the Modern Taxonomy
3 of *Oreophryne* Subfamily Asterophryinae (Anura:
4 Microhylidae)
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12 RRH: *Auparoparo* vs. *Oreophryne*
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1 ABSTRACT:

2 Despite 130 years of active study, the frog genus *Oreophryne* sensu lato
3 (Boettger 1895), has only recently been discovered as polyphyletic, consisting of two
4 unrelated reciprocally monophyletic genera. Here we show that these clades are
5 undiagnosible using morphology and provide a phylogenetic taxonomic definition for the
6 two genera. We acquired tissue samples of the type species (*O. senckenbergiana*) from
7 North Maluku Island, Indonesia, allowing for a decisive molecular phylogenetic
8 resolution for which clade retains the name *Oreophryne*, naming the other, unrelated
9 genus of arboreal frogs ***Auparoparo* gen. nov.** from the indigenous language Motu. Our
10 findings clearly demonstrate that a second clade of primarily arboreal frogs has evolved
11 within *Asterophryne*, with striking morphological similarity, given that the other non-
12 arboreal genera are quite distinct. Furthermore, our findings present a cautionary tale
13 illustrating that polyphyletic (non-evolutionary) groups may still be in use today, as
14 identifying taxa to include in phylogenetic analysis based on traditional taxonomy
15 without confirmation of monophyly, importantly, with inclusion of sister taxa (i.e.,
16 employing a phylogenetic definition) can mislead evolutionary research, even today.

17 **Key Words:** Auparoparo; frogs; homoplasy; Indo-Pacific; molecular and morphological
18 conflict; morphology; new classification; new genera; Papua New Guinea; parallelism;
19 phylogenetic nomenclature; phylogenetics; zoological nomenclature

1 ALTHOUGH the generic taxonomy of the subfamily Asterophryinae has had a
2 turbulent history (reviewed in Hill et al. 2022), the taxonomy of the genus *Oreophryne*
3 sensu lato (Boettger 1895) in particular has been comparatively stable until recently.
4 Generic designation to *Oreophryne* sensu lato has traditionally relied on a pectoral
5 anatomy that is distinct from all other genera, with highly reduced and distinctly curved
6 clavicles (Parker 1934; Zweifel 2003). Using this taxonomic definition, over the past 130
7 years the genus *Oreophryne* sensu lato (Boettger 1895) has grown to 75 diagnosed
8 species, fully 1/5 of the total diversity of the hyperdiverse subfamily Asterophryinae and
9 the largest of the ~16 genera (Hill et al. 2022; Frost 2024). In addition, other curious
10 features suggesting that *Oreophryne* sensu lato may not be a natural group include the
11 largest geographic distribution in the subfamily (Hill et al. 2023a), with species centered
12 on mainland New Guinea and its offshore islands (Werner 1898; Zweifel 1956; Zweifel
13 et al. 2003; Kraus and Allison 2004; Kraus and Allison 2009; Kraus 2013; Frost 2024),
14 but ranging westward across open ocean (Müller 1894; Boulenger 1896; Ahl 1933; Putri
15 et al. 2023) and northward to Indonesia (Boettger 1895; Setiadi et al. 2010; Günther et
16 al. 2023) and the Southern Philippines Islands (Alcala 1986; Balinton and Seña 2015).
17 Furthermore, unlike other genera that have greater degrees of lifestyle evolution,
18 *Oreophryne* sensu lato are conspicuously arboreal (or scansorial) and the genus
19 accounts for nearly all arboreal species of Asterophryinae (Günther et al. 2001; Menzies
20 2006), inhabiting lowland to montane rain forest. However, there is a small group of

1 diminutive species that inhabit high elevation grasslands, a few which are terrestrial
2 (Zweifel et al. 2005; Günther and Richards 2012; Putri et al. 2023).

3 The apparently reliable diagnostic character, however, has never been
4 demonstrated to be a synapomorphy. Recently, Hill et. al (2022, 2023b) established that
5 *Oreophryne* sensu lato represents two reciprocally monophyletic groups, temporarily
6 labeled “*Oreophryne* A” and “*Oreophryne* B”, yet it is unknown whether characters exist
7 to identify these groups. The aims of this study are to assess the phylogenetic position
8 of the newly available type species of *Oreophryne* (*Oreophryne senckenbergiana*) into a
9 large Asterophrinae (subfamily-level) phylogeny, to name and describe the new genus,
10 and to develop phylogenetic definitions for both. Furthermore, we analyze the
11 morphological characteristics to assess whether morphological diagnosis is possible,
12 and consider whether there are any additional features to assist collectors when
13 identifying putative *Oreophryne* in the field (e.g., call type, geography, other characters),
14 and discuss the evolutionary patterns and taxonomic utility of these morphological
15 characters.

16 *Taxonomic History:*

17 Boettger (1895) described the genus *Oreophryne* sensu lato and the species
18 *Oreophryne senckenbergiana* on the basis of two specimens from Halmahera Island, in
19 the Maluku Archipelago west of New Guinea, that lacked procoracoids and had
20 cartilaginous sterna. He also noted the presence of expanded finger pads and

1 transverse folds of skin in the palate. Parker (1934) noted that procoracoids were
2 present, but were reduced and re-diagnosed the genus on the basis of this feature
3 together with the presence of reduced clavicles, no omosternum and a large
4 cartilaginous sternum. Zweifel (2003) noted that “the clavicles are tiny, slightly curved
5 bones lying on (ventral to) the procoracoid cartilage and apart from the midline and the
6 scapula.” On the basis of these features there are now 75 recognized species in the
7 genus (Frost 2024).

8 While *Oreophryne* sensu lato is united morphologically ([Figure 1](#)), large scale
9 molecular phylogenetic analysis with many ingroup samples were required to clarify that
10 this group contains two reciprocally monophyletic clades (Hill et al. 2022, 2023b). Early
11 molecular phylogenetic studies suggested monophyly but only included a few taxa
12 within a larger-scale study (Köhler and Günther 2008 included 9 *Oreophryne* taxa; de
13 Sa et al. 2012 included 4). Rivera et al. (2017) provided the first molecular evidence of a
14 polyphyletic *Oreophryne* sensu lato with 36 *Oreophryne* within a larger phylogeny
15 Asterophryinae which included 155 samples, followed by Tu et al. (2018)’s sparse
16 supermatrix approach with 18 taxa of *Oreophryne* falling out in various places within a
17 much larger phylogeny, but both of these lacked strong nodal support connecting the
18 hypothesized clades to the backbone of the phylogeny. The polyphyletic nature of
19 *Oreophryne* sensu lato was not clearly demonstrated until Hill et. al (2022, 2023b)
20 included 44 taxa of *Oreophryne* with complete data sampling within a 218 taxon

1 phylogeny of Asterophryinae. This phylogeny is the most complete and robust
2 phylogeny to date. The two clades, labeled “Oreophryne A” and “Oreophryne B” have
3 deep histories of ~ 18MY and ~16MY, respectively, with “Oreophryne A” most basal to
4 the rest of the subfamily.

5 MATERIALS AND METHODS

6 Taxon Sampling

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8 Asterophryinae is an adaptive radiation with short branch lengths along the
9 backbone separating generic-level taxa (Rivera et al. 2017; Hill et al. 2022, 2023b), we
10 reconstruct the phylogeny of *Oreophryne* sensu lato within the larger context of the
11 subfamily. We include 30 species and 18 unnamed taxa of *Oreophryne* sensu lato within
12 a larger dataset of 234 samples of 203 taxa of Asterophryinae collected across New
13 Guinea, the Philippines, and Indonesia with three outgroup species to root the
14 phylogeny (*Dyscophus antongilii*, *Scaphiophryne marmorata*, and *Platypelis grandis*).
15 Importantly, we include the type species *Oreophryne senckenbergiana* (Boettger 1895)
16 in a molecular phylogeny for the first time. The samples were collected in Halmahera,
17 North Maluku Province, Indonesia by Iqbal Setiadi , [Table 1](#), near the type locality for *O.*
18 *senckenbergiana* (Halmahera, Indonesia as documented by Boettger 1895).

TABLE 1. — Locality information for samples sequenced in this study. See Hill, et al., 2023,
Table 1 for additional metadata.

ID	Species	Locality	Latitude,
			Longitude
BJE01143	<i>Oreophryne</i> <i>senckenbergiana</i>	Kabupaten Halmahera Utara,	1.7899,
		Halmahera Island, North	127.8910
		Maluku	
BJE01144	<i>Oreophryne</i> <i>senckenbergiana</i>	Kabupaten Halmahera Utara,	1.7899,
		Halmahera Island, North	127.8910
		Maluku	
BJE01120	<i>Oreophryne</i> <i>senckenbergiana</i>	Kabupaten Halmahera Utara,	1.8228,
		Halmahera Island, North	127.8300
		Maluku	

Molecular Sequence

We obtained nearly complete DNA sequence data for five loci (three nuclear: Seventh in Absentia [SIA], Brain Derived Neurotrophic Factor [BDNF], Sodium Calcium Exchange subunit-1 [NXC-1], and two mitochondrial loci (Cytochrome oxidase b [CYTB], and NADH dehydrogenase subunit 4 [ND4] from four samples of two taxa not previously sequenced and combined these with previously published data (Rivera et al. 2017; Hill et al. 2022, 2023b) for an alignment of 240 samples with 207 taxa and 2475 base pairs that is 96% complete. All loci were complete for the added samples. Sequence quality was evaluated using chromatogram quality scores, checks for stop codons indicating pseudogenes, BLAST search (Ye et al. 2012), and concordance of pairwise similarity with other Asterophryinae sequences. Primer sequences and detailed methods are provided in (Hill et al. 2023b). We note that we excluded from the previous alignment (Hill et al. 2022, 2023b) two samples of *Aphantophryne pansa* (BPBM 5299 and BPBM 8312) which were deemed unreliable as they came from formalin-fixed material, and three samples that should be confirmed with new material (BPBM 22708 *Copiula tyleri*, BPBM 38939 *Copiula sp.* 7, BPBM 37753 *Paedophryne dekot*). We note that genetic material from 47 species of *Oreophryne* sensu lato was not available for this molecular diagnosis.

Morphology, Behavior, and Geographic Distribution

We collected new morphological characters and reexamined traits compiled from the diagnosing literature , character states defined in Table 2 for all named species of *Oreophryne* in our phylogenetic dataset. Our sample included all species available in the collections of the Bishop Museum, Honolulu, Hawaii (BPBM): thirteen species of “Oreophryne A”, eight species “Oreophryne B” ([Table 3](#)). Finger and toe pads were photographed and length and width of the toe tips were measured using ImageJ software. Trait information for the remaining species were gathered from the literature, six species of “Oreophryne A” and one “Oreophryne B”. In addition, we compiled information on call type from the literature and collection location from the respective collectors or the BPBM database (global positioning coordinates [GPS]; [Table 3](#)). We examined the ability of morphological and behavioral characters to provide a diagnosis to the genus-level clade consistent with molecular phylogenetics.

TABLE 2. — Description of traits surveyed. Several traits are traditionally noted for Asterophryinae species: presence of two palatal folds, horizontal pupil, and lack of vomerine teeth (Boettger 1895; VanKampen 1923; Parker 1934).

Column*	Trait	Definition
1	Procoracoids	Size and shape of the procoracoid cartilages, CR= curved, reduced
2	Clavicles	Size and shape of the clavicles, CR= curved, reduced
3	Omasternum	Presence (+) or absence (-) of the omasternum
4	Sternum	Cartilaginous sternum or sternal plate
5	Call type	Qualitative description of the call type
6	Finger:toe width	The ratio of finger pad width to toe pad width
7	Pectoral ligament	Cartilaginous or ligamentous connection between scapula and procoracoid
8	Rel toe length	Relative length of 5th toe compared to length of 3rd toe (5<3, 5=3, 5>3)
9	Toe webbing	Toe webbing assessed on 5th toe (absent (-), basally webbed, well webbed)
10	Tympanum:eye diameter	Tympanum diameter relative to eye diameter ($\frac{1}{4}$, $\frac{1}{3}$, or $\frac{1}{2}$)
11	Reference	Citation
	Pupil horizontal	Whether the pupil is horizontal
	Teeth	Whether teeth are present

1	Palatal folds	Whether two horizontal palatal folds in front of the
2		pharynx are present

3 *Corresponding column in Table 3.

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Nomenclature

We conducted a comprehensive taxonomic literature review of diagnoses for all species known to belong to the clades “Oreophryne A” and “Oreophryne B” (Hill et al. 2022). We applied the ICZN code (Zoological Nomenclature et al. 1999) as well as the Phylocode (de Queiroz and Cantino 2020). We support the nascent effort to assign indigenous names in taxonomy whenever possible (e.g., Gillman and Wright 2020). One of us (B. Iova) is a local indigenous language expert, and provided guidance in proposing the novel genus name, *Auparoparo*, meaning “tree frog” in Motu, a regional language, for *Oreophryne* “B”.

Phylogenetic Reconstruction

We used Partitionfinder in IQTREE2 to find the best-fit evolutionary model followed by Maximum Likelihood (ML) phylogenetic reconstruction (Lanfear et al. 2012; Minh et al. 2020). We tested models partitioned by both locus and codon versus by loci only (15 vs. 5 partition scheme), and found that the locus and codon model was superior by the Akaike Information Criterion. Nodal support values for the ML tree were inferred using 2000 bootstrap replicates.

Phylogenetic Definition

We use a maximum crown-clade definition under the PhyloCode (de Queiroz and Gauthier 1990, 1992; de Queiroz and Cantino 2020), specifically, the largest crown

clade originating with the most recent common ancestor of the reference species for the clade (e.g., the type species of the genus), and all extant species of the clade more recently than they do with reference species of other clades (i.e., in this study, other generic-level clades). This definition is robust to uncertainty in basal relationships such as in the case of adaptive radiations with short branch lengths along the backbone as in Asterophryinae, as well as being robust to identification of new species and new clades.

Analyses, Visualization, and Data Availability

All analyses were conducted and graphics produced in the R Statistical Computing Environment unless noted below (R Core Team 2024). Initial DNA sequence alignment was conducted using MUSCLE (Edgar 2004) with manual adjustments in Mesquite (Maddison and Maddison 2023). Phylogenetic reconstruction was conducted using the IQTREE2 suite of software (Lanfear et al. 2012; Minh et al. 2020).

The known ranges of *Oreophryne* sensu lato were downloaded as shape files from the IUCN (2025) redlist database, and sites from collection locations for *Oreophryne* and *Auparoparo* were mapped using the sf and ggplot2 packages (Wickham 2016; Pebesma 2018), overlaid on a basemap created using rnaturalearth, grid, and ggspatial packages (Massicotte and South 2023; Dunnington 2023). Tree graphics were produced with the ggtree package Yu (2020) supplemented by ggplot2 (Wickham 2016). All data, alignments, and code to reproduce analyses are available in the public repository ...

RESULTS

Phylogeny and Phylogenetic Taxonomy

A partition scheme including codon as well as locus resulted in an improvement of fit by 3,127 AIC units (evolutionary model partitioned by locus and codon, 15-partitions: AIC of 155,492, versus partitioned by locus only, 5 partitions: AIC of 158,619). Unless otherwise noted, all phylogenetic reconstructions used the 15-partition scheme identified as best-fit by IQTREE2.

We recovered a maximum likelihood topology that confirms that *Oreophryne* sensu lato is polyphyletic, with species falling into two independently-evolved, reciprocally monophyletic, highly-supported clades [Figure 2](#). Nearly all nodes are moderately or highly supported, with most of the weakly supported nodes falling along the backbone where there are very short branch lengths as characteristic of an adaptive radiation.

Furthermore, these findings demonstrate that the clade descended from node A should retain the converted clade name (CCN) *Oreophryne*, as the type species, *O. senckenbergiana*, from North Maluku Island, Indonesia, is situated within *Oreophryne* with high support ([Figure 2](#)). We propose the new clade name (NCN) *Auparoparo* for the node descended from node B. *Oreophryne* is the oldest clade in the Papuan

subfamily, with genera *Cophixalus* and *Paedophryne* arising before *Auparoparo*, which split from the remaining genera of Asterophryinae.

Morphology, Behavior, and Ranked Taxonomy

We find no morphological synapomorphies that distinguish *Auparoparo* from *Oreophryne*. *Oreophryne* sensu lato is diagnosed by characters of the pectoral girdle: possessing highly reduced, curved procoracoid cartilages and clavicles, and lacking an omosternum (Parker 1934; Zweifel 2003). We found that this suite of characters are present in both *Oreophryne* and *Auparoparo*, and therefore is not diagnostic for neither *Oreophryne* nor *Auparoparo* ([Table 3](#)).

Two traits imperfectly segregate between the clades, toe pad widths and call type ([Table 3](#)), which may assist in preliminary identification. *Oreophryne* is an arboreal clade with expanded digital pads of roughly equal width between the fingers and toes, whereas *Auparoparo* have visibly larger finger pads, with finger pads roughly 20-30% wider than toes. Both *Oreophryne* and *Auparoparo* use a variety of call types, with “honks” or “peeps” used only by *Oreophryne*, whereas “whinny” calls are used only by arboreal *Auparoparo*. A minority of species in both groups use the “rattle” call type. Both toe pad width and call type are variable traits with some overlap and thus not reliable for diagnosis. Furthermore, toe pad is clearly an adaptive trait for arboreality.

Geographic Distribution

The geographic distributions of *Auparoparo* and *Oreophryne* are shown in [Figure 3](#), with corresponding phylogenetic relationships.

Taxonomy

We use a maximum crown-clade definition for designating *Oreophryne* and *Auparoparo* because the monophyly of these taxa are strongly supported, and the relationships of these taxa to each other and to sister genera of Asterophryinae are well-supported (this study, Hill et al. 2022, 2023b), as has been done for other anuran taxa (Brown et al. 2015) and summarized in the PhyloCode (de Queiroz and Gauthier 1990, 1992, 1994). That is, we adopt a nodal definition of the clade that identifies the genus as a monophyletic clade. Specifically, to include the type species of the genus and the most recent common ancestor that includes all extant species of the genus but not members of other genera. This definition is robust to uncertainty in basal relationships such as in the case of adaptive radiations with short branch lengths along the backbone as in Asterophryinae, as well as being robust to identification of new species. We follow Brown et al. (2015) in using the terms converted clade name (CCN) to indicate the phylogenetic redefinition of *Oreophryne*, and new clade name (NCN) for newly designated names (*Auparoparo*), whereas the previously defined polyphyletic *Oreophryne* is indicated by “sensu lato”.

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2 **Family Microhylidae (Günther 1858)**

3 **Subfamily Asterophryinae (Günther 1858)**

4 **Genus *Oreophryne* (Boettger 1895) amended**

5 *Microhyla* (Peters and Doria 1878)

6 *Callula*

7 *Oreophryne* (Boettger 1895) Type species: *Oreophryne senckenbergiana* (Boettger

8 1895). (= *Microhyla achatina* var. *moluccensis* (Peters and Doria 1878), by

9 morphological monotypy, however, this should be verified with molecular evidence.)

10 Placed on Official List of Generic Names in Zoology by Opinion 1266, Anonymous,

11 1984.

12 *Sphenophryne*

13 *Phrynixalus*

14 *Mehelyia* (Wandolleck 1911), Type species: *Meheylia lineata* (Wandolleck 1911)

15 (= *Sphenophryne biroi* (Méhely 1897), by subsequent designation of (Parker 1934).

16 Synonymy by (Nieden 1926).

17 *Cophixalus*

18 *Chaperina*

19 Type species *Oreophryne senckenbergiana* (Boettger 1895)

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Diagnosis: We note that there are no synapomorphies in adult *Oreophryne* that differ from the morphologically similar genus *Auparoparo*. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below.

Description: *Oreophryne* differs from most other genera in subfamily Asterophryinae, but not from *Auparoparo*, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape; (2) presence of well developed digital pads, often with finger and toe pads of similar size; (3) arboreal lifestyle (perch height >2m above ground); (4) most species range in body size from 16mm (*O. graminis*, Günther and Richards 2012) to 30mm, with a few larger species ~40mm (*O. inornata*); (5) small to invisible tympanum; (6) large eyes with pupil horizontal; (7) occasional presence of toe webbing; (8) thin, elongated limbs; and (9) in many species males that call with “honk” or “peep” types or in some species “rattle” type. Some species of *Oreophryne* may be distinguished from *Auparoparo* by roughly equal finger to toe pad ratio and males calling with “honk” or “peep” but not “whinny” call types.

Phylogenetic definition: *Oreophryne* (CCN) is defined as the maximum crown-clade (the monophyletic clade descending from node A, [Figure 2](#)) originating with the most recent common ancestor of *Oreophryne senckenbergiana* (Boettger 1895), the type species of *Oreophryne*, and all extant species that share a common ancestor with *O. senckenbergiana* more recently than they do with reference species of all the other

nominal generic clades listed in [Table 4](#); with reference species of the closest sister clades to *Oreophryne* currently being *Auparoparo penelopeia* (Kraus 2016), *Cophixalus verrucosus* (Boettger 1892), and *Paedophryne kathismaphlox* (Kraus 2010).

Content: Our results support the retention of 17 species in *Oreophryne* with molecular evidence ([Table 5](#)): *Oreophryne anamiatoi* (Kraus and Allison 2009), *O. anulata* (Stejneger 1908), *Oreophryne aurora* (Kraus 2016), *Oreophryne biroi* (Méhely 1897), *Oreophryne brachypus* (Werner 1898), *Oreophryne cameroni* (Kraus 2013), *Oreophryne geislerorum* (Boettger 1892), *Oreophryne graminis* [Günther and Richards (2012)], *Oreophryne inornata* (Zweifel 1956), *Oreophryne lemur* (Kraus 2016), *Oreophryne lorae* (Boulenger 1898), *Oreophryne nana* (Brown and Alcala 1967), *Oreophryne notata* (Zweifel 2003), *Oreophryne parkeri* (Loveridge 1955), *Oreophryne parkopanorum* (Kraus 2013), *Oreophryne philosylleptoris* (Kraus 2016), and *Oreophryne senckenbergiana* (Boettger 1895). A number of species were not available for our study but should molecular evidence demonstrate their relationship within *Oreophryne*, those species should also retain their generic identity.

Distribution: The range of confirmed species in this dataset remains nearly as broad as previously defined ([Figure 3](#)), from the Philippines and Indonesia to the southeastern most tip of mainland New Guinea and several near-shore islands (Fergusson, and Normanby Is.) and the distant Rossel Island of the Louisiade Archipelago. However, the transfer of species which occupy Woodlark Islands and

Misima and Sudest Islands of the Louisiade Archipelago reduces the known range of *Oreophryne*. Woodlark Island is inhabited by *O. phoebe* [= *Auparoparo phoebe*], Misima Island is inhabited by *O. picticus* [= *Auparoparo picticus*], and Sudest Island is inhabited by *O. ezra* and *O. meliades* [= *Auparoparo ezra* and *Auparoparo meliades*].

Family Microhylidae (Günther 1858)

Subfamily Asterophryinae (Günther 1858)

Genus *Auparoparo* gen. nov.

Oreophryne (Boettger 1895)

Sphenophryne (anthonyi) (Boulenger 1897)

Type species *Oreophryne penelopeia* (Kraus 2016) [= *Auparoparo penelopeia*]

Diagnosis: We note that there are no synapomorphies in adult *Auparoparo* that differ from the morphologically similar genus *Oreophryne*. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below.

Description: *Auparoparo* differs from most other genera in subfamily Asterophryinae, but not from *Oreophryne*, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape; (2) presence of well developed digital discs, with finger discs notably wider than toe discs (typically 20% or more); (3) arboreal lifestyle (perch height >2m above ground); (4) small to moderately large body size in known species, from 16mm (*A. brunnea*, Kraus

2017) to 25 mm (*A. picticus*, Kraus 2016); (5) small to invisible tympanum; (6) large eyes with pupil horizontal; (7) occasional presence of toe webbing; (8) and thin, elongated limbs; and (9) and in many species males that call with a “whinny” call type or in some species “rattle” type. Some species of *Auparoparo* may be distinguished from *Oreophryne* by larger finger pads than toe pads and males calling with “whinny” but not “honk” or “peep” call types.

Phylogenetic definition: *Auparoparo* (NCN) applies to the largest crown clade originating in (CCN) is defined as the maximum crown-clade originating with the most recent common ancestor of *Auparoparo penelopeia* (Kraus 2016), the type species of *Auparoparo*, and all extant species that share a common ancestor with *A. penelopeia* more recently than they do with the type species of all the other nominal genera listed in [Table 4](#), the type species of the closest sister taxa being *Oreophryne senckenbergiana* (Boettger 1895), *Cophixalus verrucosus* (Boettger 1892), and *Paedophryne kathismaphlox* (Kraus 2010).

Content: We transfer eleven named species from *Oreophryne* into *Auparoparo* ([Table 5](#)): *Auparoparo brunnea* (Kraus 2017), *Auparoparo equus* (Kraus 2016), *Auparoparo ezra* (Kraus and Allison 2009), *Auparoparo insulana* (Zweifel 1956), *Auparoparo matawan* (Kraus 2016), *Auparoparo meliades* (Kraus 2016), *Auparoparo penelopeia* (Kraus 2016), *Auparoparo phoebe* (Kraus 2017), *Auparoparo picticus*

(Kraus 2016), *Auparoparo nana*, *Auparoparo penelopia* (Kraus 2016), *Auparoparo variabilis* (Boulenger 1896).

Distribution: Southeastern tip of mainland Papua New Guinea and satellite islands (Fergusson, Misima, Normanby, Rossel, Sudest, Woodlark Is.), Sulawesi.

Etymology: The generic name *Auparoparo* comes from the indigenous language, Motu, which is widely spoken in the region of Papua New Guinea in which this genus is found. It is a combination of the words “au” (tree) and “paro paro” (frog). Motu is not a gendered language, however, for nomenclatural purposes we designate the genus *Auparoparo* as feminine, thereby preserving the gender of existing species names.

DISCUSSION

Our phylogenetic results (Hill et al. 2022; 2023b, this study) provide clarity on the classification of the problematic genus *Oreophryne* sensu lato. We confirm that *Oreophryne* sensu lato is polyphyletic and consists of two reciprocally monophyletic clades, “Oreophryne A” and “B” (of Hill et al. 2022), originating 18 and 16 MYA, respectively ([Figure 2](#)). Here we designate “Oreophryne A” as retaining the name *Oreophryne* based on its containing the type species *Oreophryne senckenbergiana* (Boettger 1895) with high support. *Oreophryne* is the earliest diverged genus in subfamily Asterophryinae, sister to all the other genera. The second clade, “Oreophryne B”, is highly supported as a separate, unrelated genus, which contains no known

1 species with historical generic priority. Thus, we designate this genus as *Auparoparo*, a
2 nominal tribute to the Motu language of the indigenous peoples of the region.

3 The Difficulty with Cryptic Genera

4 Perhaps the greatest testament to the crypticism of *Oreophryne* (CCN)
5 vs. *Auparoparo* (NCN) is that no taxonomist was able to distinguish that they are
6 separate clades, independently evolved, in the 130 years between their discovery and
7 phylogenetic classification [Boettger (1895); this study]. These species are indeed
8 morphologically very similar ([Figure 1](#)) and our results provide some clues as to why
9 *Oreophryne* taxonomy has been so difficult. We found no morphological synapomorphy
10 for neither *Auparoparo* nor *Oreophryne*, and moreover, no morphological feature that
11 can diagnose either clade or both in combination.

12 The traditional basis for generic designation of species to *Oreophryne* has
13 primarily relied on a signature pectoral anatomy (whether original or revised diagnosis,
14 Parker 1934; Zweifel 2003). Our phylogenetic results clearly show that this trait appears
15 more than once, in both *Oreophryne* and *Auparoparo*, and rather than being a highly
16 conserved trait, it is either a symplesiomorphy (shared ancestral character) or a result of
17 convergent or parallel evolution, and perhaps adaptive for arboreality. Thus, the pectoral
18 anatomy is not a reliable diagnostic character in differentiating between *Oreophryne* and
19 *Auparoparo*. In studies across a wide diversity of anurans, Soliz et al. (2025) found that
20 anatomical variation of the pectoral girdle corresponds with functional and ecological
21 lifestyle demands, whereas Engelkes et al. (2020) found strong phylogenetic influence

1 in the shape of the pectoral girdle. This point would be excellent for further study within
2 Asterophryinae, as there are high elevation terrestrial species of *Oreophryne* sensu lato
3 that possess the same pectoral trait (Zweifel et al. 2005), as well as a few arboreal
4 species within other genera (*Cophixalus*, *Asterophrys*, *Xenorhina*) with unknown
5 pectoral morphology.

6 We further reexamined additional characters previously used to diagnose
7 *Oreophryne* as well as characters conventionally used to diagnose species of frogs. We
8 conclude that pectoral girdle connector type (Parker 1934) and relative lengths of the
9 3rd and 5th toe are uninformative and the degree of toe webbing is subjective. It was
10 previously suggested that species in *Oreophryne* sensu lato could be divided into two
11 groups based on the type of connection between the procoracoid cartilages and the
12 scapula (Kraus 2013), with roughly half possessing a cartilaginous rod extending to the
13 scapula and others a ligamentous connection (Parker 1934). However, our phylogenetic
14 evidence shows that these traits readily evolve and are not synapomorphies and
15 therefore not reliable indicators of monophyletic groups. Ligamentous connectors are
16 common across both genera ([Table 3](#)). Of species with toe webbing, *Oreophryne* tends
17 to be described as having “well webbed” toes compared to *Auparoparo*’s “basal”
18 webbing ([Table 3](#)). However, intra-specific variability combined with the subjective
19 nature of these descriptions and inconsistency between specific markers when used
20 (“reaching the penultimate tubercle”, “webbed 1/3 length of 5th toe”, etc.), makes this

feature difficult to standardize for generic diagnostic use. Many of the morphological traits proposed by Parker (1934) and Zweifel (2003) may in fact be a result of homoplasy to evolutionary pressures imposed by lifestyle or parallel evolution, and while they may prove useful for ecomorphological identification or species diagnoses in combination with other traits, are not informative for generic diagnosis.

We found only two characters with potential for imperfectly identifying genera: relative digital pad size (Parker 1934; Zweifel 1956), and call type (e.g., Kraus 2016). We found that the relative widths of the digit pads differs between genera: *Auparoparo* tends to have visibly larger finger pads in comparison to toe pads, whereas in *Oreophryne* the finger and toe pads are roughly equal in size ([Figure 1](#), bottom row), although there is some overlap in finger to toe pad ratio ([Table 3](#)). In some cases, *Auparoparo*'s finger pads display T-shapes while the toes are often rounder and narrower. Whether there is any performance advantage in larger finger pads compared to toes is unknown. Furthermore, a handful of high-elevation species not included in our phylogeny have evolved terrestrial or fossorial habits and possess smaller digital pads compared to their presumed arboreal relatives (Zweifel et al. 2005).

Call features are increasingly used for diagnosing species of *Oreophryne*, with advertisement call types commonly categorized as rattle, honk, whinny, peeping, or clicking (Kraus 2016). We found honk or peep call types are only found in *Oreophryne* species in our sample, while whinny calls are only found in *Auparoparo* species, and

1 rattle calls are found in both genera ([Table 3](#)). We note that several authors described
2 species of each genus, thus the difference between genera cannot be attributed to the
3 investigator. However, more complete sampling may add further complexity. As far as
4 we are aware there has been no comprehensive acoustic analysis across *Oreophryne*
5 sensu lato. More research is needed to explore call diversity both between species and
6 across genera to understand call evolution and whether there are any features that are
7 distinctive of genera.

8 **Geographic Distribution**

9 Both *Oreophryne* and *Auparoparo* are geographically centered in the mainland
10 New Guinea. *Oreophryne* extends southward to the Louisiade Archipelago and
11 northward to the South Philippine Islands (Hill et al. 2023b). Addition of the type species
12 *Oreophryne senkenbergiana* expands the known range eastward to North Maluku,
13 Indonesia ([Figure 3](#)).

14 The known distribution of *Auparoparo* is mainly clustered in the southeastern tip
15 of mainland Papua New Guinea (in the East Papuan Composite Terrane) and satellite
16 islands with two species in Sulawesi, Indonesia. This suggests the existence of yet
17 undescribed or undiscovered species of *Auparoparo* with intermediate localities. This
18 geographic range is sure to grow with additional phylogenetic sampling. For example
19 *Oreophryne* (sensu lato) *chlorops* (Günther et al. 2023) is a very large (SVL > 40mm)
20 species recently discovered from mid-elevation mountains of the bird's neck region of
21

1 Western Papua with clearly enlarged finger pads relative to toe pads. If confirmed as
2 *Auparoparo*, it would significantly expand the range across New Guinea Island.

3 We found geographic range overlap between *Oreophryne* and *Auparoparo* at
4 multiple localities: on the mainland along the Mt. Dayman Foothills, at the Mt. Simpson
5 Massif, Mt. Trafalgar, and on the islands of Normanby and Fergusson (both nearshore
6 islands), and Rossel island (hundreds of kilometers away), which presents interesting
7 questions regarding their ecological relationships. For example, *Oreophryne inornata*
8 and *Auparoparo insulana* have been collected at the same site (Goodenough Island)
9 and even the same microhabitats (Zweifel 1956). How can species coexist in the same
10 niche? And what is the temporal history of each colonization? Are both genera also
11 present in the nearby islands of Normanby and Fergusson (all part of the
12 D'Entrecasteaux Archipelago)? Obtaining additional geographical samples for inclusion
13 in phylogenetic analysis should be a high priority to gain a more complete
14 understanding of the geographic distribution and evolutionary history of both
15 *Oreophryne* and *Auparoparo*.

16 Phylogenetic Taxonomy

17 We adopt an explicitly phylogenetic framework for the new taxonomy of the
18 genera *Oreophryne* and *Auparoparo*, and to our knowledge, this is the first time that a
19 phylogenetic taxonomic approach has been applied for any genus of the subfamily
20 Asterophryinae. A nodal definition of genera is advantageous especially because
21

1 Asterophryinae is an adaptive radiation. While we are increasingly confident that at least
2 16 genera are indeed monophyletic, nevertheless, very short branch lengths separate
3 many of the genera along the backbone [Hill et al. (2022); this study]. A nodal definition
4 is robust to uncertainty (i.e., rearrangements of order) along the backbone. We define
5 genera as the largest crown clade originating with the most recent common ancestor of
6 the reference species for the clade (e.g., the type species *Oreophryne senkenbergiana*),
7 and all extant species of the clade more recently than they do with reference species of
8 other clades that the sample may potentially belong to. The type species, if genetic
9 material is available seems like a reasonable choice for the reference taxon, but for
10 groups such as Asterophryinae, where a number of type species were collected in the
11 1800's it may not be possible to obtain a genetic sample, as these locations may be
12 uncertain and/or remote, and other proxies may be needed. Nevertheless, is
13 increasingly clear that for *Oreophryne* and *Auparoparo*, a molecular phylogenetic
14 approach is required for final confirmation of taxonomy.

15 Furthermore, a phylogenetic definition provides guidance for robust classification.
16 Given the deep histories of the two genera, it is imperative that molecular confirmation
17 of either *Oreophryne* or *Auparoparo* include members of the other clade as an outgroup,
18 but also members of *Cophixalus*, *Paedophryne*, or other closest sister genera as
19 outgroups to clearly resolve whether the sample shares an MRCA more closely with
20 *Oreophryne* or *Auparoparo*. The ability to resolve the “boundaries” of the monophyletic

1 groups requires it. An additional difficulty here is that within each genus, there are
2 subclades with deep histories, so care must be taken to decisively demonstrate
3 monophyly. For example, a phylogenetic analysis with only *Oreophryne* sensu lato and
4 outgroups to Asterophryinae, will show grouping of *Oreophryne* and *Auparoparo*,
5 because of lack of intervening phylogenetic history. Similarly, an analysis with only
6 *Oreophryne* sensu lato with a member of one of the younger genera, say, *Hylophorbus*,
7 will give a similar misleading result because both *Oreophryne* and *Auparoparo* are
8 equally distant from *Hylophorbus*. Clearly a greater taxonomic representation is always
9 better, but in this case, care is required to avoid a misleading result. This study
10 furthermore highlights the importance of taxonomic sampling for robust phylogenetic
11 results. Developing an orderly taxonomy in a clade with over 350 named species (and
12 an estimated 40% more yet to be described species currently in museum collections)
13 would be that much more challenging without the benefit of generic diagnoses in this
14 interesting group.

15 Clarifying the generic structure also facilitates appreciation for the patterns of
16 diversification: a tendency toward niche conservatism within genera, ecological and
17 generic diversification occurring early in the history of the subfamily (Hill et al. 2022);
18 and furthermore the very interesting biogeographical history of simultaneous dispersal
19 of multiple genera to offshore islands (Hill et al. 2022, 2023a) revealing the very strong
20 influence of plate tectonics for explaining the historical dispersal patterns of this group.

1 Finally, without stability in the naming of genera and the independent histories
2 embodied by each, it becomes much more difficult to appreciate that the two clades
3 represented by *Oreophyne* and *Auparoparo* have independently evolved, that there are
4 two independent arboreal clades, and would make it very difficult to appreciate the
5 impressive convergence or parallelism embodied by these speciose and successful
6 genera. An accurate and informative taxonomy serves to advance our understanding of
7 this group and the very interesting evolutionary insights that they embody leading to
8 further discovery.

9 Lastly, we emphasize that future naming of species, genera, and higher
10 classifications should always seek to recognize the language of the indigenous
11 stewards of the land and ecosystem from which they were discovered (Gillman and
12 Wright 2020). This is especially true of Asterophryinae frogs, whose awe-inspiring
13 diversity mirrors the rich cultural beauty and historical complexity of Papuan people.

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DATA AVAILABILITY

The data underlying this article are available in the GenBank Nucleotide Database] at <https://www.ncbi.nlm.nih.gov/genbank/> , and can be accessed with [will be provided upon publication]. The codes and datafiles needed to reproduce these results can be found at [GitHub repository] with DOI [supplied upon publication].

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FIGURE CAPTIONS

FIG. 1.—Portraits in life of arboreal frogs in Asterophryinae. *Oreophryne* species on the left, *Auparoparo* on the right, species names as labeled. Comparison of finger and toe pad sizes in *Oreophryne inornata* BPBM16217 vs. *Auparoparo equus* BPBM15775, bottom row.

FIG. 2.—Molecular phylogenetic hypothesis of Asterophryinae showing polyphyly of *Oreophryne* sensu lato, into two reciprocally monophyletic clades, node A is *Oreophryne* (blue clade) and node B is *Auparoparo* (red clade) based on the position of the type species *Oreophryne senkenbergiana*. Molecular evidence includes two mitochondrial and three nuclear markers analyzed using Maximum Likelihood following Rivera et al. (2017); Hill et al. (2022). Nodal support: black dots indicate ML bootstrap support $\geq 95\%$, grey dots $\geq 75\%$, numeric values given for nodes below 75% support. Outgroup taxa excluded.

FIG. 3.—The known distribution of *Oreophryne sensu lato* is plotted in grey polygons (data source IUCN 2025), with points indicating *Auparoparo* (red) and *Oreophryne* (blue) as identified by our phylogeny Figure 2. Note that there is overlap among points